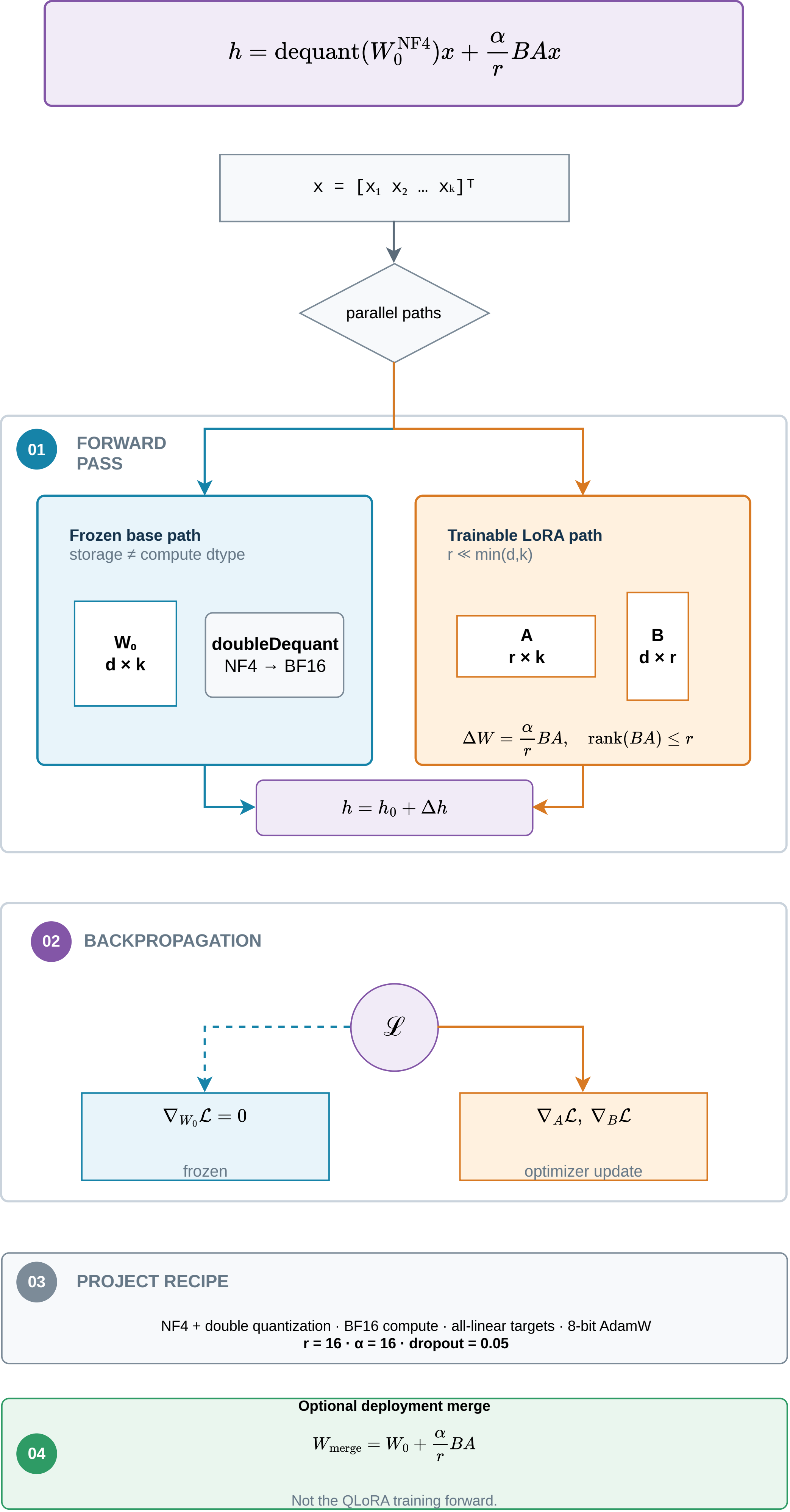
$h = \mathrm{dequant}(W_0^{\mathrm{NF4}})x + \frac{\alpha}{r}BAx$
x = [x_1 x_2 … x_k]^T
parallel paths
01
FORWARD PASS
Frozen base path
storage ≠ compute dtype
W_0
d × k
doubleDequant
NF4 → BF16
Trainable LoRA path
r ≪ min(d,k)
A
r × k
B
d × r
$\Delta W = \frac{\alpha}{r}BA, \quad \mathrm{rank}(BA) \leq r$
$h = h_0 + \Delta h$
02
BACKPROPAGATION
$\mathscr{L}$
$\nabla_{W_0}\mathcal{L} = 0$
frozen
$\nabla_A\mathcal{L},\ \nabla_B\mathcal{L}$
optimizer update
03
PROJECT RECIPE
NF4 + double quantization · BF16 compute · all-linear targets · 8-bit AdamW
r = 16 · α = 16 · dropout = 0.05
Optional deployment merge
04
$W_{\mathrm{merge}} = W_0 + \frac{\alpha}{r}BA$
Not the QLoRA training forward.